Austin Moon

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Education

University of Waterloo Waterloo, ON
B.A.Sc. Mechatronics Engineering, Specialization in Artificial Intelligence *Sep. 2024 – Expected June 2029*

Technical Skills

Languages: Python, C/C++, TypeScript, CSS, SQL, VHDL, MATLAB
AI/ML & CV: PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, NumPy, pandas, Matplotlib
Embedded/Tools: STM32, Arduino, I2C/SPI/UART, CAN bus, Git, Linux, VS Code
Certificates: Machine Learning Specialization – Supervised/Unsupervised/Reinforcement Learning

Experience

Machine Learning Engineer Sep 2026 – Present
WAT.ai – OffBall XI *Waterloo, ON*
- Working on reinforcement learning (Google Research Football) and graph-based models (PyTorch Geometric) to simulate counterfactual off-ball movement and decision-making from professional soccer tracking data

Tech Executive – AI/ML, Full-Stack Engineer Sep 2026 – Present
UW Association of Korean-Canadian Scientists and Engineers *Waterloo, ON*
- Building an AI resume tailoring app that classifies job postings and scores resume bullets with Jev, rewrites them with LLM APIs using field specific prompts, and generates tailored resumes

Embedded Software Developer May 2025 – Aug 2026
Waterloo Aerial Robotics Group *Waterloo, ON*
- Developed a C++ driver for a three-axis magnetometer on STM32, using I2C communication with DMA to read magnetic flux density and compute a real-time compass heading
- Extracted and parsed position, altitude, and speed from a u-blox M10 GPS receiver through an interrupt-driven UART NMEA stream, converting the data into navigation coordinates
- Implemented a low-pass filter in C for the ICM-42688 IMU within WARG's firmware, reducing noise in sensor data

Mechatronics Engineering Intern Sep 2025 – Dec 2025
Spyder Controls Corporation *Lacombe, AB*
- Developed a Python script for an Arduino Nano to validate thermistors by comparing analog voltage readings and flagging resistance values outside a ±5% tolerance, open circuits, and short circuits
- Flashed and validated firmware and circuits on 10+ products, systems, and test jigs over a CAN bus network

Projects

NFL Defensive Coverage Predictor | *Python, PyTorch, ByteTrack, YOLOv11, OpenCV* | *GitHub*
- Trained a custom YOLOv11 object detection model on a football helmet dataset with 10,000+ images, achieving 69% mAP50 and 75% precision, and integrated ByteTrack for multi-object tracking across occlusions
- Engineered 339-dimensional feature vectors from raw NFL tracking data (4.8M rows, 18 weeks, 11.5K plays) and trained a PyTorch MLP to classify 6 defensive coverage types using player positioning and formation indicators
- Built an end-to-end video-to-prediction inference pipeline with automated field calibration via Hough line transforms for yard-line and hash-mark detection, coordinate mapping, and snap-frame detection

NBA Game Predictor | *Python, NumPy, pandas, scikit-learn* | *GitHub*
- Built a Python ETL pipeline that used a REST API to retrieve NBA team and player statistics for the 2025–26 season, then cleaned, validated, and exported the datasets to CSV using pandas for downstream modeling
- Engineered rolling-average features and compared logistic regression, random forest, and gradient boosting classifiers on 1,000+ games, achieving 63% accuracy (vs. approximately 54% majority-class baseline)
- Implemented a Tkinter dashboard for selecting dates and generating game-specific predictions

G-Helper Clone | *Python, PyQt6, D-Bus* | *GitHub*
- Developed a Python and PyQt6 system tray application for managing ASUS laptop hardware on Fedora Linux, leveraging Claude AI as a development assistant throughout the build
- Tuned 50+ kernel and hardware parameters (CPU EPP, PCIe ASPM, PCI runtime PM), reducing idle power by 80%
- Built a debounced state machine to auto-configure CPU frequency, GPU DPM, and fan PWM on battery power
- Engineered a real-time hwmon sensor system with Tdie/Tctl readings and battery discharge calculations